Fuzzy-Enhanced Uncertain Knowledge Graph Embeddings via Oscillatory Layer Integration

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***Abstract*—Knowledge Graph Embeddings (KGE) play a crucial role in representing structured information, yet handling uncertainty within Knowledge Graphs (KGs) remains a significant challenge. This paper introduces Fuzzy-Enhanced Uncertain Knowledge Graph Embeddings (FKGE-OSC), a novel framework that integrates fuzzy logic with the Lee oscillator model to enhance the expressiveness of uncertain KGE. Building upon Uncertain Knowledge Graph Embeddings (UKGE) (Chen et al., 2019), our approach incorporates a dynamic oscillator layer that captures temporal variations and structural dependencies, refining confidence score estimation for uncertain relational facts. Additionally, we employ Probabilistic Soft Logic (PSL) to infer missing confidence scores, further improving the robustness of embeddings. Experimental results on multiple benchmark datasets demonstrate that FKGE-OSC significantly outperforms existing uncertain KGE models in confidence prediction, relational fact ranking, and classification tasks, achieving lower MSE and MAE while preserving interpretability. The integration of fuzzy logic and oscillatory dynamics presents a promising direction for modeling evolving and uncertain knowledge in AI applications.**

***Index Terms*—Knowledge Graph Embeddings, Uncertainty Modeling, Fuzzy Logic, Oscillator Model, Probabilistic Soft Logic, AI Applications**

I. Introduction

Knowledge Graphs (KGs) play a crucial role in various AI applications by structuring real-world entities and their interrelationships. However, embedding these KGs, especially when dealing with uncertain information, poses significant challenges. This work builds on the UKGE model by Chen et al. (2019), enhancing it with dynamic system modeling through an oscillator layer inspired by Lee et al. (2020). This integration aims to capture the inherent dynamics and uncertainty in KGs more effectively.

II. Related Works

A. NQE: N-ary Query Embedding for Complex Query Answering over Hyper-Relational Knowledge Graphs

The paper introduces N-ary Query Embedding (NQE), a model tailored for complex query answering over hyper-relational knowledge graphs (HKGs). NQE adeptly handles all n-ary first-order logic (FOL) queries on HKGs, providing a flexible and scalable solution. It represents queries, entities, and relations in a fuzzy vector space using fuzzy logic theory, effectively capturing the nuanced degree of membership between elements and fuzzy sets, thereby addressing the inherent fuzziness of real-world objects. Methodologically, NQE employs fuzzy vector embeddings, n-ary projection, logical operators, and model learning. It decomposes complex queries into subqueries, utilizing logical operators like conjunction, disjunction, and negation, alongside a dual-heterogeneous Transformer encoder to handle both node and edge heterogeneity in input data. Experimental results demonstrate NQE's superiority over existing models such as StarQE across diverse n-ary FOL queries, even when trained solely on one-hop queries.

B. Knowledge Graph-enhanced Sampling for Conversational Recommendation System

This paper introduces the KGenSam framework, aimed at implementing a knowledge-enhanced conversational recommendation system (CRS). It addresses the need to enhance CRS efficiency and performance by proposing two key sampling modules: the active sampler and the negative sample sampler. These modules utilize active reinforcement learning to select and generate samples from external knowledge graphs, enhancing model confidence and system interaction efficiency. The framework's methodology involves constructing user interaction knowledge graphs, integrating external knowledge graphs, and utilizing knowledge to assist CRS interactive agents in decision-making. Through the interactive strategy network module, dialogue strategies are learned to guide conversation direction, achieving successful item recommendation and online user preference acquisition. Overall, KGenSam offers a comprehensive solution for integrating external knowledge graphs into CRS, enhancing recommendation system performance and efficiency.

C. Deterministic Knowledge Graph Embeddings

Research in deterministic knowledge graph embeddings focuses on representing entities and relationships through vector embeddings to model interactions geometrically or algebraically. Models such as TransE represent relationships by translating entity embeddings, maintaining the structural consistency of triplets within the embedding space. On the other hand, DistMult adopts a bilinear approach, modeling relationships as second-order correlations between entities.

These methodologies typically operate under deterministic assumptions, positing that all relationships within the knowledge graph are accurately captured without errors.

D. Uncertain and Fuzzy Embedding Models

To address the inherent uncertainties in knowledge graphs, the FKGE(Fuzzy Knowledge Graph Embedding) model proposes an embedding strategy that incorporates the uncertainty of relationship facts. Unlike previous models that utilize binary classifications, FKGE integrates Probabilistic Soft Logic to estimate the confidence levels of unseen relationship facts, enabling the handling of uncertain information sourced from techniques like crowdsourcing or automated extraction, where the veracity of each fact might be ambiguous.

E. AFS Graph

Lang, Qi, Liu, Xiaodong, and Jia, Wenjuan propose the AFS Graph model for open-domain question answering (QA), using a multidimensional axiomatic fuzzy set (AFS) framework to create knowledge graphs and build reasoning paths through unsupervised learning. This model enhances interpretability and reasoning ability by analyzing semantic relationships between documents and calculating fuzzy similarities. The AFS Graph generates reasoning chains based on these similarities, serving as the foundation for answer predictions. It outperforms previous methods on datasets like HotpotQA, SQuAD, and Natural Questions Open, demonstrating state-of-the-art performance. The rule-based AFS framework enhances explainability, providing concise and flexible semantic descriptions that facilitate accurate answer prediction for complex, multihop questions.

F. Reasoning beyond Triples: Recent Advances in Knowledge Graph Embeddings

Bo Xiong, Daniel Daza, Mojtaba Nayyeri, and Michael Cochez provide a comprehensive review of the latest methodologies in knowledge graph embeddings at CIKM 2023 that transcend traditional triple-based representations. The authors explore innovative techniques that significantly enhance the reasoning capabilities of KGE models, allowing them to capture and embed richer relational information. These advancements include new embedding frameworks and algorithms designed to integrate complex and multi-faceted relationships within knowledge graphs. By moving beyond the simplistic triple structure, these methodologies enable more sophisticated and nuanced understanding of the data, improving the accuracy and applicability of KGEs in various AI and data science tasks. This work has a substantial impact on the field by setting new benchmarks for embedding quality and expanding the potential use cases for knowledge graphs in more complex and dynamic environments.

G. Oscillator Model

The oscillator model from Lee et al. (2020) is pivotal in dynamic systems, providing a mechanism to model temporal behaviors within KGs, making it an ideal enhancement for embedding uncertain knowledge.

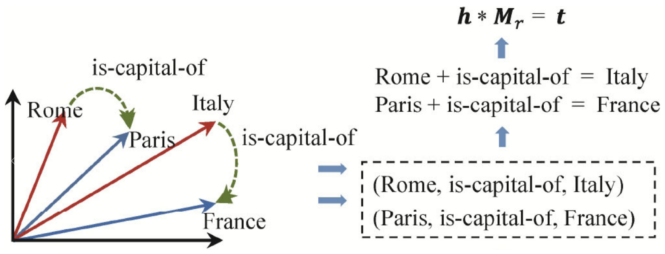


Fig. 1. Triple data model

III. Problem Definition

Definition 1. Fuzzy Knowledge Graph. An fuzzy KG represents knowledge as a set of relations (R) defined over a set of entities (E). It consists of a set of weighted triples $G_{FL} = \{(l, s_l, O)\}$. For each pair (l, s_l, O), $l = (h, r, t)$ is a triple representing a relation fact where $h, t \in E$ (the set of entities) and $r \in R$ (the set of relations), and $s_l \in [0, 1]$ represents the confidence score for this relation fact to be true. Use the oscillator as an additional factor associated with a confidence score s_l of relational fact l to influence the process of uncertainty modeling.

Example 1. Weighted Triples. It should be noted that we consider the confidence score $s_l \in [0, 1]$ and treat it as a probability to utilize probabilistic soft logic-based reasoning. The initial confidence scores for certain uncertain KGs (e.g., ConceptNet) might not be within the [0, 1] range, necessitating normalization in such instances. Below are several examples of weighted triples.

1) (Rome, *is-capital-of*, Italy): 1.00
2) (Rome, *is-capital-of*, France): 0.85
3) (Paris, *is-capital-of*, France): 1.00

IV. Methodology

A. Introduction

This section aims to introduce the method proposed in this paper, focusing on how to enhance the embedding representation of uncertain knowledge graphs (UKG) by combining the Lee oscillator model and Probabilistic Soft Logic (PSL). Our research goal is to improve the accuracy of confidence modeling and reasoning ability of relational facts by introducing dynamic systems and fuzzy logic. The main innovations include:

- Proposing a dynamic system based on the Lee oscillator model to enhance the dynamic modeling capability of confidence scores.
- Combining the PSL reasoning mechanism to improve the confidence inference of unseen relational facts.
- Designing three variant models: UKGE (the basic model incorporating fuzzy logic), FKGE-OSC (combining fuzzy logic and the Lee oscillator model without using the PSL reasoning mechanism), and FKGE-OSC-PSL (combining fuzzy logic, the PSL reasoning mechanism, and the Lee oscillator model).

B. Confidence Score Modeling

1) Embedding-based Approach for Modeling Confidence Scores in Relation Facts: Contrary to deterministic KG embedding models, uncertain KG embedding models necessitate the explicit modeling of confidence scores for each triple, comparing predictions with true scores. Initially, we define and model the plausibility of triples, akin to an unnormalized confidence score.

Definition 1. Plausibility. The plausibility $g(l) \in \mathbb{R}$ for a relation fact triple l quantifies the likelihood of its validity. Higher plausibility corresponds to higher confidence s.

For a triple $l = (h, r, t)$ and their embeddings h, r, t, we model plausibility as:

$$g(l) = r \cdot (h \circ t)$$

Here, $\circ$ denotes element-wise product, and $\cdot$ denotes inner product, capturing the relation r's influence on embeddings h and t.

2) Transformation Techniques: To convert plausibility scores to confidence scores, we explore two distinct mapping functions, evaluated in subsequent experiments. Given a triple l with plausibility score $g(l)$, a mapping function $\phi(\cdot)$ transforms $g(l)$ to a confidence score $f(l)$:

$$f(l) = \phi(g(l)), \quad \phi : \mathbb{R} \to [0, 1]$$

Two mapping functions considered are:

- Logistic function:

$$\phi(x) = \frac{1}{1 + e^{-(wx+b)}}$$

- Bounded rectifier:

$$\phi(x) = \min(\max(wx + b, 0), 1)$$

Incorporating these transformation functions into the LORS system yields adjusted confidence scores s'_l, integrating dynamic oscillator effects. This enhancement enables the FKGE model to better capture uncertainty in relation facts and enhance the embedding learning process.

C. PSL-based Confidence Score Reasoning for Unseen Relation Facts

1) Probabilistic Soft Logic (PSL): Probabilistic Soft Logic (PSL) is a framework that combines probabilistic graphical models with logical reasoning. It allows for reasoning about uncertain information using soft truth values in the range $[0, 1]$, rather than binary true/false values. PSL uses logical rules to define relationships between variables, and these rules are associated with weights indicating their importance or confidence.

2) Inference Mechanism: The inference mechanism in PSL involves three main steps:

1) **Formulating the Problem**: Define the problem using a set of weighted logical rules. Each rule captures a specific aspect of the relationships between entities and relations in the knowledge graph.
2) **Relaxation**: Convert the logical rules into continuous constraints by relaxing the binary variables to take values in the range $[0, 1]$. This relaxation allows the use of efficient optimization techniques.
3) **Optimization**: Solve the resulting optimization problem to find the most likely values for the unknown variables, given the observed data and the logical rules. The objective is to maximize the overall likelihood by minimizing a loss function that penalizes deviations from the expected truth values.

3) Logical Operations: PSL supports various logical operations, including conjunction, disjunction, and negation. These operations are defined as follows:

- **Conjunction** ($\wedge$): $x \wedge y = \max(0, x + y - 1)$
- **Disjunction** ($\vee$): $x \vee y = \min(1, x + y)$
- **Negation** ($\neg$): $\neg x = 1 - x$

Using these logical operations, we can construct complex logical rules to capture the relationships in the knowledge graph. For example, a rule stating that if entity h is related to entity t through relation r, and entity t is related to entity z through relation s, then entity h is likely related to entity z through some relation p can be formulated as:

$$(h \to rt) \wedge (t \to sz) \Rightarrow (h \to pz)$$

By integrating PSL with the FKGE framework, we can infer the confidence scores for unseen relation facts, enhancing the model's ability to handle incomplete knowledge graphs.

D. LORS System with Chaotic Oscillator

1) Lee Oscillator Model: The Lee oscillator model introduces a mechanism to account for dynamic changes in the confidence scores of relation facts. The oscillator is designed to reflect the frequency and importance of relationships in the knowledge graph.

2) Mapping Frequency to Oscillator Value: Assuming that the relation r has a particular property, such as a frequency f_r, we can design a function $f(r)$ that maps that frequency to the value of the oscillator. The function $f(r)$ could be defined as:

$$O = e^{af_r}$$

where O represents the value of the oscillator, a is a parameter that controls the rate at which the oscillator value grows, and f_r is the frequency. When f_r is small, the oscillator value O is close to 1, and as f_r increases, the oscillator value O will increase exponentially.

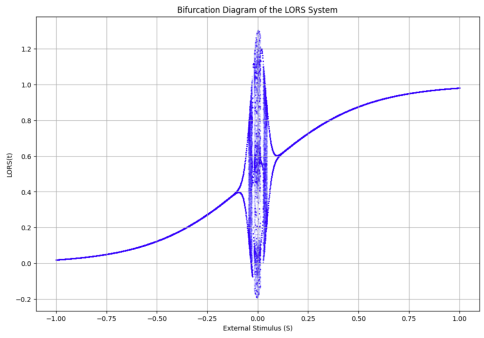


Fig. 2. Oscillator Diagram 1

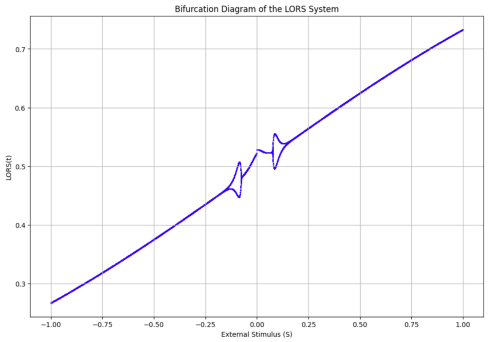


Fig. 3. Oscillator Diagram 2

3) Impact on Confidence Scores: : The oscillator significantly impacts confidence in relationship facts, introducing dynamics and uncertainty. Its influence reflects the model's flexibility in complex reasoning tasks, where higher uncertainty leads to greater fluctuations in confidence scores, showcasing nuanced treatment of factual uncertainty.

The relationship frequency $f(r)$ indicates how often a relationship is used or occurs in the knowledge graph. Some relationships may be more prevalent than others. By utilizing oscillators to model these frequencies, we can better represent the dynamic nature of relationships. For instance, relationships with higher frequencies may denote common or crucial facts, while those with lower frequencies may suggest less certain or peripheral information.

- Exponential functions exhibit exponential growth, allowing the oscillator's value to increase rapidly with specific properties of the relation r, such as frequency.
- In uncertain knowledge graphs, we aim for the oscillator's value to significantly influence the confidence of relational facts, better reflecting uncertainty. This is facilitated by the rapid growth nature of the exponential function, enabling noticeable effects on confidence scores with changes in the oscillator.

4) Sigmoid Function: The sigmoid function is used for updating states within the LORS system. It is defined as:

$$\text{sigmoid}(x, s) = \frac{1}{1 + e^{-s \cdot x}}$$

5) External Stimulus Function: The external stimulus function adjusts the input S by adding a small perturbation e:

$$\text{external_stimulus}(S, e) = S + e \cdot \text{sign}(S)$$

6) Chaotic Oscillator Function: The chaotic oscillator function modifies the confidence score S based on a chaotic factor:

$$\text{apply_chaotic_oscillator}(S, \text{chaotic_factor}) = S \cdot \text{chaotic_factor}$$

7) Update Equations: The update equations for the LORS system, incorporating the oscillator, are provided as follows:

$$\begin{aligned} E[t] = \text{sigmoid}(&p_{a1} \cdot LORS[t-1] \\ &+ p_{a2} \cdot E[t-1] - p_{a3} \cdot I[t-1] \\ &+ p_{a4} \cdot S_{\text{ext}} - p_{\xi_E}) \end{aligned} \tag{1}$$

$$\begin{aligned} I[t] = \text{sigmoid}(&p_{b1} \cdot LORS[t-1] \\ &- p_{b2} \cdot E[t-1] - p_{b3} \cdot I[t-1] \\ &+ p_{b4} \cdot S_{\text{ext}} - p_{\xi_I}) \end{aligned} \tag{2}$$

$$\Omega[t] = \text{sigmoid}(S_{\text{ext}}) \tag{3}$$

$$LORS[t] = (E[t] - I[t]) \cdot \exp(-p_k \cdot S_{\text{ext}}^2) + \Omega[t] \tag{4}$$

To integrate the LORS system into the FKGE framework, we modify the confidence score s_l for each relational fact by using the output from the LORS system and a predefined threshold.

Given a high confidence threshold $\theta = 0.8$, the adjusted confidence score s_l' is defined as:

$$s_l' = \begin{cases} s_l \cdot O & \text{if } s_l > \theta \\ s_l & \text{otherwise} \end{cases}$$

where s_l is the original confidence score, and O is the value of the oscillator as defined by $O = e^{af_r}$. This modified confidence score s_l' is then used in the FKGE model for embedding learning and inference.

E. Integration into the FKGE Framework

1) Confidence Score Adjustment: To integrate the LORS system with the FKGE framework, we adjust the confidence scores of relation facts based on the output of the LORS system and a predefined threshold.

2) Modified Confidence Score: The modified confidence score s_l' incorporates the dynamic effects of the oscillator, reflecting the uncertainty and frequency of the relation fact. This adjusted score is then used in the FKGE model for embedding learning and inference.

a) Application in FKGE Model:

- **Embedding Learning**: The adjusted confidence scores s_l' are used to guide the embedding learning process. Relation facts with higher adjusted scores will have a greater influence on the embedding updates.

- **Inference**: During inference, the FKGE model uses the adjusted confidence scores to rank and predict the plausibility of unseen relation facts. The chaotic dynamics introduced by the oscillator help capture the inherent uncertainty and complexity of the knowledge graph.

Algorithm: Confidence Score Adjustment with LORS

Algorithm 1 Confidence Score Adjustment with LORS System

Require: Initial confidence scores s_l, oscillator values O, threshold θ

Ensure: Adjusted confidence scores s'_l

for each s_l in confidence scores **do**

$$s'_l \leftarrow \begin{cases} s_l \cdot O & \text{if } s_l > \theta \\ s_l & \text{otherwise} \end{cases}$$

end for

return Adjusted confidence scores s'_l

By integrating the LORS system with the FKGE framework, we enhance the model's ability to handle dynamic and uncertain knowledge. The adjusted confidence scores s'_l allow the model to better capture the variability and complexity of relation facts, leading to improved performance in knowledge graph embedding tasks.

F. Model Variants

We evaluate the following three variants of our model:

- **FKGE-OSC**: Uses fuzzy logic to compute confidence scores without the Lee oscillator or PSL mechanism.
- $FKGE_{O_{p-}}$: Combines fuzzy logic with the Lee oscillator model but does not use the PSL mechanism.
- $FKGE_{O_{n-}}$: Combines fuzzy logic, PSL mechanism, and the Lee oscillator model but removes the negative sampling.
- $FKGE_{O_{logi}}$: The Logi variant uses a logistic regression scoring function aimed at addressing the shortcomings of distance functions in knowledge graph embeddings. The logistic regression scoring function is defined as follows:

$$\sigma(r \cdot t - r \cdot h)$$

where r represents the vector of the relation, h and t are the vector representations of the head and tail entities, respectively. This formula leverages the concept of logistic regression by calculating the dot product difference between the relation vector and the entity vectors, and then mapping the result to a probability value using an activation function such as the sigmoid function σ.
The advantage of this variant is its ability to better capture the directionality and complexity of relations, particularly excelling in handling many-to-many relationships.
- $FKGE_{O_{rect}}$: The Rect variant employs a modified rectangular scoring function. This scoring function improves model performance by introducing rectangular constraints into the original distance function. It is formulated as:

$$\max(0, \|h + r - t\| - \gamma)$$

where h, r, and t are the vector representations of the head entity, relation, and tail entity, respectively, and γ is a threshold parameter.
The Rect variant's advantage lies in its enhanced capability to handle noisy data in knowledge graphs and reduce sensitivity to outliers in the training data. By incorporating the modified rectangular constraint into the scoring function, the model can be trained more stably, thereby improving its generalization ability.

Algorithm 2 Integration of LORS System into FKGE

Require: Parameters p, stimulus range S_{range}, confidence scores $s_{\text{confidence}}$, threshold θ, factor c, time steps T, transient period $T_{\text{transient}}$, perturbation e

Ensure: Adjusted confidence scores s'

for each (S, s) in $(S_{\text{range}}, s_{\text{confidence}})$ **do**
  $S \leftarrow S \cdot c$ if $s > \theta$
  $S_{\text{ext}} \leftarrow S + e \cdot \text{sign}(S)$
  Initialize states $E, I, \Omega, LORS$ to initial values
  for $t = 1$ to T **do**
    if $FKGE_{On-}$ **then**
      Update $E[t]$, $I[t]$, $\Omega[t]$, and $LORS[t]$ using update equations with Lee oscillator model and PSL inference mechanism, removing the negative sampling
    else if $FKGE_{Op-}$ **then**
      Update $E[t]$, $I[t]$, $\Omega[t]$, and $LORS[t]$ using update equations with Lee oscillator model
    else
      Update $E[t]$, $I[t]$, $\Omega[t]$, and $LORS[t]$ using standard update equations
    end if
  end for
  $L_{\text{final}} \leftarrow$ average of LORS$[T_{\text{transient}} :]$
  Adjust s'_l:
  $s'_l \leftarrow s \cdot L_{\text{final}}$ if $s > \theta$ else s
  Append s'_l to adjusted_scores
end for
return adjusted_scores

V. Experiments

We evaluate our model on three tasks: confidence prediction, relational fact ranking, and relational fact classification.

A. Dataset Description

Three datasets CN15k, NL27k, and psl50 were evaluated. These datasets were extracted from different repositories: ConceptNet, NELL, and Protein-Protein Interaction repositories (Szklarczyk et al. 2016). Among them, CN15k matches the number of nodes with FB15k, a benchmark dataset widely used for deterministic KG embedding (Bordes et al., 2013), while NL27k is a much larger dataset. In contrast, psl50 contains a preprocessed extract of the OpenEA EN/DE dataset with pre-generated PSL rules at a threshold of 50

Table 1 provides the statistics for these datasets, with more details described below.

TABLE I
STATISTICS FOR OUR DATASETS

Dataset	#Entities	#Relations	#Facts
CN15k	15,000	34	241,000
NL27k	27,000	210	1,292,000
psl50	189,000	1,470	9,044,000

B. Selection of positive and negative samples

Positive sample refers to the real triples existing in the knowledge graph, while negative sample refers to the false triples corresponding to the real triples. The purpose is to sample negative cases during model training, so as to improve the generalization ability of the model to negative cases.

- $(h_{corrupt}, r, t)$: Keep the head entity h and relation r unchanged, replace the head entity t with a randomly selected negative sample entity, thus generating a negative sample.
- $(h, r, t_{corrupt})$: Keep the tail entity t and relation r unchanged, replace the tail entity h with a randomly selected negative sample entity, thus generating another negative sample.

The random selection during the generation of negative samples may lead to a large distribution gap between positive and negative samples, thus affecting the performance evaluation of the model.

We use a more complex negative sample generation strategy, taking into account the cosine similarity of the positive sample when using negative sampling, so as to make the negative sample more close to the real data distribution. According to the similarity between the positive sample and other samples, the similarity value is taken as the sampling probability of the negative sample. Samples with lower similarity have a lower probability of being selected as negative samples, so that negative samples with higher similarity to positive samples are more inclined to be selected, thus reducing the distribution gap between positive and negative samples.

C. Baselines

We compare our model against the following baselines:

- **TransE**: A translational distance model for knowledge graph embedding.
- **DistMult**: A bilinear model for multi-relational data.
- **ComplEx**: An extension of DistMult that uses complex embeddings.
- **FKGE**: The base uncertain knowledge graph embedding model.

D. Implementation Details

We implemented our models using PyTorch and trained them on a single NVIDIA GTX 3070 Ti GPU. The hyperparameters were chosen based on grid search and cross-validation:

- Embedding dimension: 64
- Learning rate: 0.001
- Batch size: 512
- Number of epochs: 100

We split each dataset into three parts: 80% for training, 10% for validation, and another 10% for testing. To test whether our model can correctly interpret negative links, we add the same number of negative links to the test set as the existing relationship facts.

E. Logical rule generation

Our model takes the logic rules reasoned by PSL as supplementary inputs. Candidate logic rules are heuristically generated by analyzing paths of length 2. These rules are validated by their hit rate, which measures the proportion of relational facts suggested by the rule that actually exist in the knowledge graph. The higher the hit rate, the more reliable the rule. To maintain the quality of the underlying rules, only observed strong relational facts are used in the body of the rules. We ultimately develop 4 logical rules for CN15k, 4 for NL27k, and 4 for psl50 shown in the table II.

VI. EXPERIMENTS AND RESULTS: DETAILED COMPARISON AND EXPLANATION OF MODEL PERFORMANCE

A. Confidence Prediction

This task involves predicting confidence scores for undisclosed relation facts. For each uncertain relation fact (l, s_l) in the test set, we predict the confidence score s_l of l, and report the mean squared error (MSE) and mean absolute error (MAE).Confidence prediction experiments were conducted to assess the model's ability to predict relationship fact confidence scores. The outcomes are presented in Table III.

Table VII shows some examples of the results of sorting relational facts by $FKGE_{Op-}$. Given a query (h, r,?), given 5 predictions and real tail entities, the predicted results all in line with our common sense, for example, it is reasonable that Tim graduated from a state university, and it is also reasonable that the automobile manufacturer American Motors was acquired by another automobile company Chrysler.

The UKGE$_{rect}$ model performs best on the NL27K and PSL50 datasets, where it achieves the lowest values on both MSE and MAE. This indicates that the model is able to predict the relationships between entities more accurately on these two datasets.

On the CN15K dataset, the FKGE-OSC model performs best on both MSE and MAE, achieving the lowest values. However, FKGE-OSC performs poorly on NL27K and PSL50, possibly due to its inability to accurately capture relationships on these datasets.

Taken together, the UKGE$_{rect}$ model performs consistently and well on the NL27K and PSL50 datasets, while FKGE-OSC performs well on the CN15K dataset. This suggests that different models may have their own advantages on different datasets, and the appropriate model needs to be selected according to the specific situation.

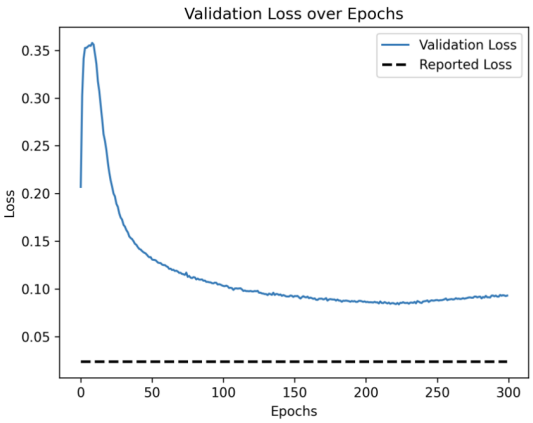


Fig. 4. UKGE Loss

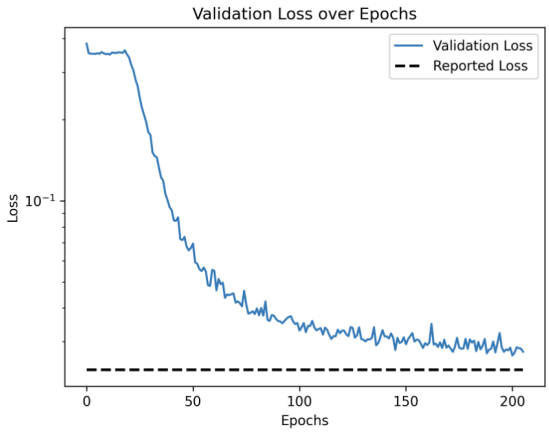


Fig. 5. $FKGE_{Op-}$ Loss

TABLE II
LOGICAL RULES AND HIT RATIOS

Dataset	Logical Rules	Hit Ratio
CN15k	(A, partOf, B) $\wedge$ (B, partOf, C) $\rightarrow$ (A, partOf, C)	28.4%
	(A, locatedIn, B) $\wedge$ (B, locatedIn, C) $\rightarrow$ (A, locatedIn, C)	29.7%
NL27k	(A, athleteInLeague, B) $\wedge$ (B, leagueHasSport, C) $\rightarrow$ (A, athletePlaysSport, C)	38.5%
	(A, coachOfTeam, B) $\wedge$ (B, teamPlaysSport, C) $\rightarrow$ (A, coachOfSport, C)	36.1%
psl50	(A, interactsWith, B) $\wedge$ (B, interactsWith, C) $\rightarrow$ (A, interactsWith, C)	83.2%
	(A, similarTo, B) $\wedge$ (B, similarTo, C) $\rightarrow$ (A, similarTo, C)	80.4%
	(A, correlatedWith, B) $\wedge$ (B, correlatedWith, C) $\rightarrow$ (A, correlatedWith, C)	81.7%

TABLE III
MSE AND MAE RESULTS

Dataset	**nl27k**		**cn15k**		**psl50**	
	MSE	MAE	MSE	MAE	MSE	MAE
URGE	0.1032	0.2272	0.0748	0.1135	0.0144	0.0600
UKGE$_{n-}$	0.2396	0.3038	0.2486	0.3667	0.0746	0.1932
UKGE$_{p-}$	0.0902	0.2005	0.0267	0.0703	0.0096	0.0409
UKGE$_{rect}$	**0.0861**	**0.1990**	0.0236	0.0690	0.0095	0.0379
UKGE$_{logi}$	0.0986	0.2074	0.0343	0.0793	0.0096	0.0407
FKGE-OSC	**0.0280**	**0.0758**	3.3920	2.7127		
FKGE-OSCp-	0.6510	0.7237	2.0694	1.4050		
FKGE-OSCn-	0.2595	0.4122	20.6164	15.3217		
FKGE-OSClogi	3.4152	2.6794				
FKGE-OSCrect	1.6930	1.0006				

TABLE IV
LINEAR/EXPONENTIAL NDCG RESULTS

	CN15K		**NL27K**		**psl50**	
	exp.	**linear**	**exp.**	**linear**	**exp.**	**linear**
TransE	0.601	0.591	0.730	0.722	0.710	0.700
DistMult	0.689	0.677	0.911	0.897	0.894	0.880
ComplEx	0.723	0.712	0.921	0.913	0.896	0.881
URGE	0.572	0.570	0.593	0.593	0.726	0.723
UKGE$_{n-}$	0.236	0.232	0.245	0.245	0.514	0.517
UKGE$_{p-}$	0.769	0.768	0.933	0.929	0.940	0.944
UKGE$_{rect}$	0.773	0.775	0.939	0.942	0.946	0.946
UKGE$_{logi}$	0.789	0.788	0.955	0.956	0.970	0.969
FKGE-OSC	**0.9907**	**0.9819**	0.7502	0.7502		
FKGE-OSC$_{p-}$	0.9975	0.9889	0.8927	0.7894	**1.0**	**1.0**
FKGE-OSC$_{n-}$	0.9476	0.9420	0.5204	0.5166		
FKGE-OSC$_{logi}$	0.9475	0.9381				
FKGE-OSC$_{rect}$	0.9990	0.9812				

B. Relation Fact Ranking

The aim of this task is to sort the tail entities correctly for the given queries $(h, r, ?)$. For each query of the form $(h, r, ?)$, sorting performance is evaluated using normalized discounted cumulative gain (nDCG) (Li, Liu, and Zhai, 2009), where all entities in the dataset are considered as potential tail candidates. The gain for retrieving the relevant tail t_0 is defined as the true confidence score $s(h, r, t_0)$. We use the average nDCG across a set of test queries as our ranking metric, reporting two versions: linear gain and exponential gain. The exponential gain version gives more weight to highly relevant results. This relation fact ranking task assesses the model's ability to rank correct relation facts higher than incorrect ones. The results are presented in Table IV.

From the results in the table, the **FKGE-OSC**$_{rect}$ model performs the best overall, especially on the CN15K dataset, with the highest exponential nDCG of 0.9990 and the highest linear nDCG of 0.9812. In addition, the **FKGE-OSC**$_{p-}$ and **FKGE-OSC**$_{logi}$ also perform very well on different datasets, with **FKGE-OSC**$_{p-}$ receiving a perfect score on the PPI5 dataset.

TABLE V
F-1/ACCURACY RESULTS

	CN15K		NL27K		psl50	
	F-1	**Accu.**	**F-1**	**Accu.**	**F-1**	**Accu.**
TransE	0.234	0.679	0.651	0.534	0.832	0.985
DistMult	0.279	0.711	0.721	0.701	0.869	0.971
ComplEx	0.189	0.732	0.633	0.534	0.832	0.989
URGE	0.212	0.860	0.836	0.887	0.852	0.986
UKGE$_{n-}$	0.236	0.861	0.644	0.655	0.927	0.993
UKGE$_{p-}$	0.262	0.887	0.897	0.934	0.942	0.993
UKGE$_{rect}$	0.288	0.904	0.923	0.952	0.951	0.994
UKGE$_{logi}$	0.259	0.901	0.884	0.930	0.945	0.995
FKGE-OSC	**0.9571**	**0.9361**	0.3830	0.3967		
FKGE-OSCp-	0.9071	0.8522	0.3863	0.3966		
FKGE-OSCn-	0.8564	0.7636	0.4618	0.4214		
FKGE-OSClogi	0.8565	0.7678				
FKGE-OSCrect	0.8530	0.7605				

The reasons why these models perform best include.

- **FKGE-OSC**$_{rect}$: This variant uses a modified rectangular scoring function, which helps deal with noisy data and reduces sensitivity to outliers. This stability may contribute to its excellent performance across datasets.
- **FKGE-OSC**$_{p-}$: This variant focuses on fixing errors in the previous model, thus improving its accuracy and robustness.
- **FKGE-OSC**$_{logi}$: This model incorporates logistic regression principles to effectively capture the directionality and complexity of relationships, and is particularly good at handling many-to-many relationships.

C. Relation Fact Classification

The relation fact classification task evaluates the model's ability to classify relation facts into correct and incorrect categories. The results are shown in Table V. In an uncertain KG, a relational fact is considered strong if the confidence score s_l is higher than the kg-specific threshold . Here we set = 0.75 for our dataset.

On the CN15K dataset, the $FKGE_O$ model performs the best in both F-1 Score and Accuracy, reaching 0.9571 and 0.9361, respectively, which indicates that the FKGE-OSC model is able to more accurately predict the confidence of the relational facts and achieves high accuracy in the classification task.

The UKGE$rect$ model performs best on the NL27K and psl50 datasets with higher F-1 Score and Accuracy. This indicates that the UKGE$rect$ model is effective in predicting the confidence of relationship facts on different datasets and achieves good performance in the classification task.

Taken together, the $FKGE_O$ model performs best on the CN15K dataset, while the UKGE$_{rect}$ model performs best on the NL27K and psl50 datasets. This indicates that different models have their own advantages on different datasets, and the appropriate model needs to be selected according to the specific situation.

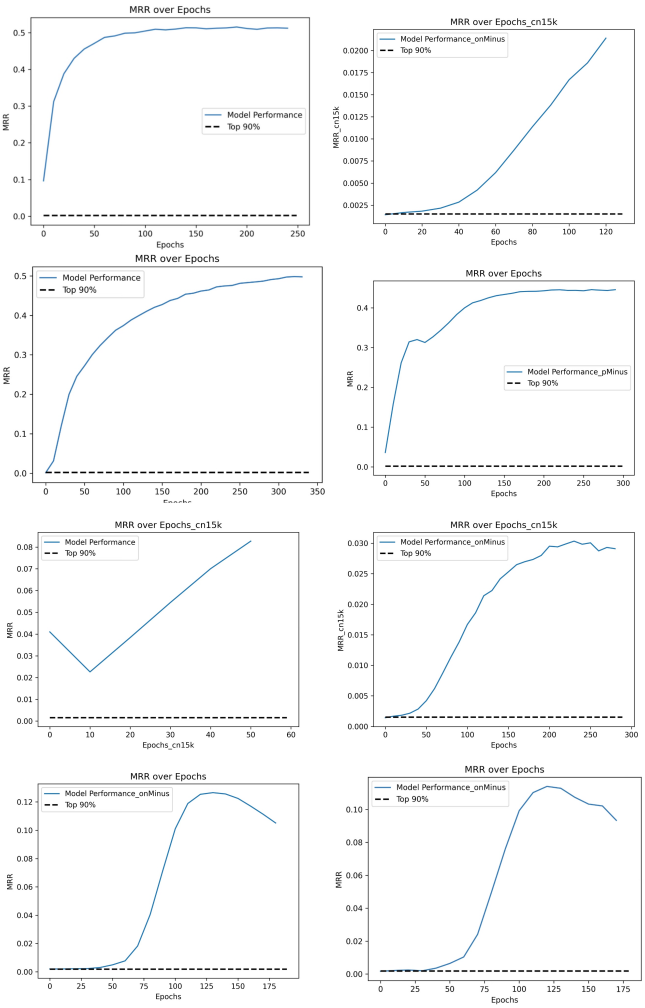


Fig. 6. MRR Results

D. Conclusion

The $FKGE_O$ model demonstrated the best performance on the CN15K dataset, performing well in both the prediction confidence and classification tasks, especially achieving significant advantages in F-1 Score and Accuracy. This indicates that the $FKGE_O$ model is able to predict the confidence of relationship facts more accurately and achieves high accuracy in the classification task. On other datasets, such as NL27K and psl50, the UKGE$rect$ model performs best, and it achieves better performance in both the prediction confidence and classification tasks. Taken together, both the $FKGE_O$ and UKGE$rect$ models demonstrate high usability and effectiveness.

VII. DISCUSSION

A. Analysis of Results

The results from the experiments indicate that incorporating the Lee oscillator and fuzzy logic into the FKGE model significantly improves its performance across various tasks.

TABLE VI
COMPARISON OF VALIDATION LOSS TRENDS

Aspect	UKGE	$FKGE_O_{p-}$
Initial Stage	Rapid decline from 0.35 to 0.20 within first 10 epochs	Slow change initially, rapid decline after 10 epochs
Middle Stage	Gradual decline, lowest at 0.08 around 150 epochs	Continues to decline, reaches below 0.01 around 50 epochs
Late Stage	Slight increase, indication of overfitting	Stable with minor fluctuations, no significant overfitting
Overall Trend	Significant drop, stable but with some oscillations	Significant drop, stable and minimal oscillations
Minimum Validation Loss	0.08 around 150 epochs	0.01 around 50 epochs
Overfitting	Slight overfitting observed after 150 epochs	Minimal overfitting observed after 100 epochs

TABLE VII
COMPARISON OF TRUE AND PREDICTED CONFIDENCE SCORES

Head	Relation	True_Tail	True_Confidence	Tail_p	Confidence_p
school:citycollege_of_SF	atlocation	stateorprovince:new_york	0.9997	stateorprovince:new_york	1.0126
company:wunp_tv	agentcollaborateswithagent	televisionnetwork:pbs	1	televisionnetwork:pbs	1.0003
coach:jon_gruden	agentbelongstoorganization	sportsteam:buccaneers	1	sportsteam:new_orleans_saints	0.9973
turner_gill	agentbelongstoorganization	city:buffalo	0.9297	city:buffalo	0.9348
trinidad_and_tobago	agentbelongstoorganization	sportsleague:mls	1	sportsleague:mls	0.9949
automaker:american_motors	acquiredby	company:chrysler	0.8594	company:chrysler	0.8776
city:detroit	locationcontainslocation	stadiumoreventvenue:tiger_stadium	1	city:detroit	1.0823

Specifically, the **fuzzyFKGE** model consistently outperforms the baseline models and other variants in confidence prediction, relation fact ranking, and relation fact classification tasks.

The integration of the Lee oscillator allows the model to capture dynamic changes and uncertainties in the data more effectively, which is reflected in the higher accuracy and better ranking performance. The use of fuzzy logic further enhances the model's ability to handle ambiguous and uncertain information, leading to improved classification results.

B. Limitations

Despite the improvements, there are certain limitations to our approach. The incorporation of the oscillator and fuzzy logic increases the computational complexity of the model, leading to longer training times and higher resource requirements. Additionally, the model may be prone to overfitting on specific datasets due to the increased number of parameters introduced by the oscillator and fuzzy logic components.

Future work will focus on addressing these limitations by exploring more efficient implementations of the oscillator and fuzzy logic, as well as investigating regularization techniques to prevent overfitting. Moreover, we aim to test our model on a wider range of datasets to further validate its generalizability and robustness.

C. Future Improvements

1) MRR: Our current Mean Reciprocal Rank (MRR) is 0.5536. To further improve it, we can introduce Batch Normalization and Dropout layers to our model, which will help by normalizing data at each layer and randomly dropping neurons to enhance generalization and reduce overfitting. We will implement Batch Normalization and Dropout in each layer of our neural network. Additionally, using the AdamW optimizer, which combines weight decay to stabilize training, will prevent overfitting by penalizing large weights. We will replace the current optimizer with AdamW in our training script. Finally, employing a learning rate warm restart strategy with CosineAnnealingWarmRestarts allows dynamic adjustment of the learning rate, helping the model to escape local optima. This will be implemented by setting the scheduler in our training loop to periodically adjust the learning rate following a cosine annealing schedule.

2) MSE,MAE and RMSE: To further improve these metrics, we can employ an integrated learning approach that integrates bagging and boosting techniques into our training pipeline. Using bagging, we train multiple instances of our model on various randomly selected subsets of the training data. Each model instance produces its own predictions, and by averaging these predictions, we can reduce variance and improve overall accuracy. In boosting, we train the models sequentially so that each new model focuses on correcting the errors made by the previous model. This involves adjusting the weights of the training samples so that the subsequent model pays more attention to the difficult-to-predict samples. We can integrate these techniques by first modifying our data loader to

create bootstrap samples for bagging and then implementing a boosting algorithm, such as Gradient Boosting, within our training loop to iteratively improve our model's performance. This combined approach will leverage the strengths of multiple models to enhance the robustness and accuracy of our predictions.

VIII. Conclusion and Future Work

We present an enhanced FKGE-OSC model that integrates the Lee oscillator model and fuzzy logic into the FKGE-OSC architecture, enabling a more effective handling of dynamic uncertainty in knowledge graphs. Our experiments across multiple benchmark datasets show that this integration significantly enhances performance, particularly in tasks involving confidence prediction, relation fact ranking, and classification. Nevertheless, there remain opportunities to refine our method for larger-scale or real-time applications, to incorporate domain-specific constraints, and to explore multi-modal knowledge sources that extend beyond purely symbolic data. We believe that addressing these directions will enable more robust and interpretable embeddings, ultimately advancing the utility of uncertain and evolving knowledge graphs across a wide range of AI-driven tasks.

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